

Data Source

File: `Marketing_Campaign_Raw_Data.csv` — a synthetic dataset simulating one year of marketing campaigns for a fictional Indian e-commerce company, **UrbanKart**, across 5 channels.

Data Description

220 campaign records, 10 columns. Channels: Email, Google Ads, Social Media, TV, Radio.

Column	Description
Campaign_ID	Unique identifier for each campaign
Campaign_Name	Campaign title (channel + occasion/sale event)
Channel	Marketing channel used
Start_Date / End_Date	Campaign run dates
Budget_Spent (INR)	Amount spent on the campaign
Impressions	Number of times the campaign was shown
Clicks	Number of clicks received
Conversions	Number of completed purchases attributed to the campaign
Revenue_Generated (INR)	Revenue attributed to the campaign

Table 1: Raw dataset structure

10 of 220 rows had at least one missing value: Impressions (2), Clicks (1), Conversions (3), Revenue_Generated (4). No duplicate Campaign_IDs, no date/budget formatting issues.

Data Cleaning Process

1. Converted `Start_Date` and `End_Date` from text to proper date values.
2. Added a `Duration (Days)` column: `End_Date - Start_Date`.
3. Handled missing values (see Section 5).
4. Added 6 calculated columns (see Section 6).
5. Color-coded every corrected, estimated, or flagged cell (see Section 4).

Color-Coding Convention

Three colors were used across the cleaned sheet:

Color	Meaning
Red	Negative ROI / loss-making campaign
Grey	Missing value that could not be estimated, left blank (Revenue, and the ROI/Profit-Loss cells depending on it)
Yellow	Missing value calculated from other fields (Impressions, Clicks, Conversions)

Table 2: Color-coding legend

Campaign_ID	Conversions	Revenue_Generated (INR)	ROI (%)
C044	161.93	16,310,979	90.48
C003	3,263	—	—
C093	128	176,210	−0.45

Table 3: Example rows

Handling Missing Values Using Other Available Data

- **Missing Impressions / Clicks (3 rows)** — calculated using the channel’s average CTR:

$$\text{Impressions} = \frac{\text{Clicks}}{\text{Average CTR}}, \quad \text{Clicks} = \text{Impressions} \times \text{Average CTR}$$

- **Missing Conversions (3 rows)** — calculated using average Conversion Rate:

$$\text{Conversions} = \text{Clicks} \times \text{Average Conversion Rate}$$

- **Missing Revenue (4 rows)** — not estimated. Revenue has no stable ratio to spend/clicks, so guessing it would distort ROI. Left blank; ROI and Profit/Loss also left blank for these rows.

Derived Columns Added

Six new columns were added:

Column	Formula
Duration (Days)	End_Date − Start_Date
CTR (%)	Clicks / Impressions
Conversion Rate (%)	Conversions / Clicks
Cost Per Click	Budget_Spent / Clicks
Cost Per Conversion	Budget_Spent / Conversions
ROI (%)	(Revenue_Generated − Budget_Spent) / Budget_Spent × 100
Profit/Loss	Revenue_Generated − Budget_Spent

Table 4: Derived metrics added during cleaning

Channel Performance Overview

Channel	Budget (INR)	Revenue (INR)	Overall ROI (%)
Email	14,64,045	11,13,24,460	75.04
Google Ads	1,11,97,820	37,67,34,075	32.64
Social Media	84,77,891	16,38,59,311	18.33
TV	1,91,34,047	25,344,413	0.32
Radio	68,97,985	69,62,828	0.01

Table 5: Channel-wise budget, revenue, and ROI

Best Performing Channel

Email has the best ROI at 75% overall (81% average per campaign), well ahead of every other channel. It is also the cheapest to run — lowest cost per click (Rs. 2.44) and lowest cost per conversion (Rs. 44.58) — with the highest CTR (4.3%) and conversion rate (6.0%). Low cost plus high conversion is why it wins on ROI.

Campaigns To Discontinue

29 campaigns are running at a loss. All 29 are TV or Radio — 18 of 40 TV campaigns and 11 of 16 Radio campaigns. Not a single Email, Google Ads, or Social Media campaign is loss-making. The worst campaign, C193 (TV), costs Rs. 7,104 per conversion — over 150x Email’s average.

Recommendation: pause or drop the loss-making TV and Radio campaigns, and review these two channels before running new campaigns on them.

Where To Increase Budget

Email is under-funded — only 3% of total spend, despite the best ROI. Google Ads is the biggest revenue driver (Rs. 37.7Cr) with solid ROI (33%) too.

Recommendation: shift budget out of TV (40% of total spend, near-zero ROI) and Radio, into Email and Google Ads.

Surprising Findings

- TV gets the largest budget of any channel but the second-worst ROI. High reach (26.5 crore impressions) does not translate into conversions.
- Every single loss-making campaign is TV or Radio — none from the other three channels. Channel choice, not campaign execution, is driving the losses.
- C211 posted an outlier ROI of 436%, well above the next-best campaign (243%) — worth a manual check.
- 4 campaigns have no revenue recorded at all, a data gap worth flagging upstream.

Summary of Recommendations

1. Increase budget for Email — best ROI, currently smallest allocation.
2. Keep scaling Google Ads — largest revenue driver with solid ROI.
3. Cut the 29 loss-making TV and Radio campaigns.
4. Reduce overall TV and Radio spend and reinvest in Email and Google Ads.
5. Investigate the C211 outlier and the 4 missing-revenue campaigns before relying on totals that include them.